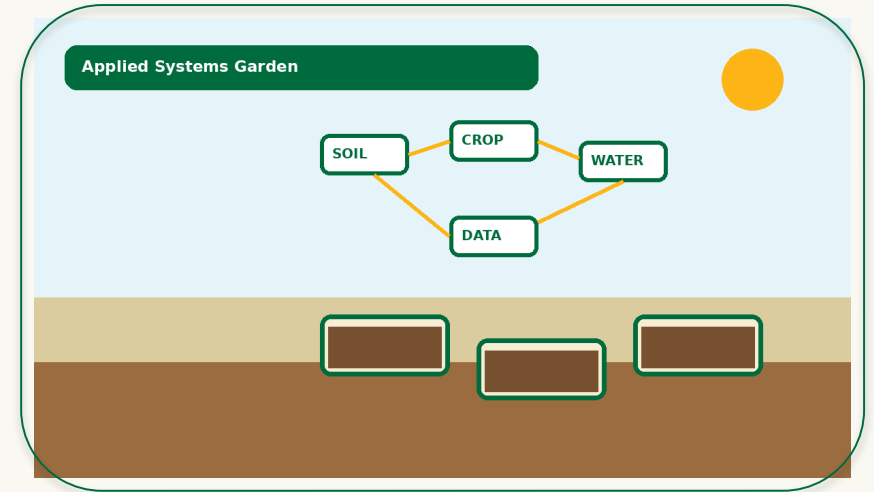


Applied Systems Garden

Connecting Agricultural Subsystems

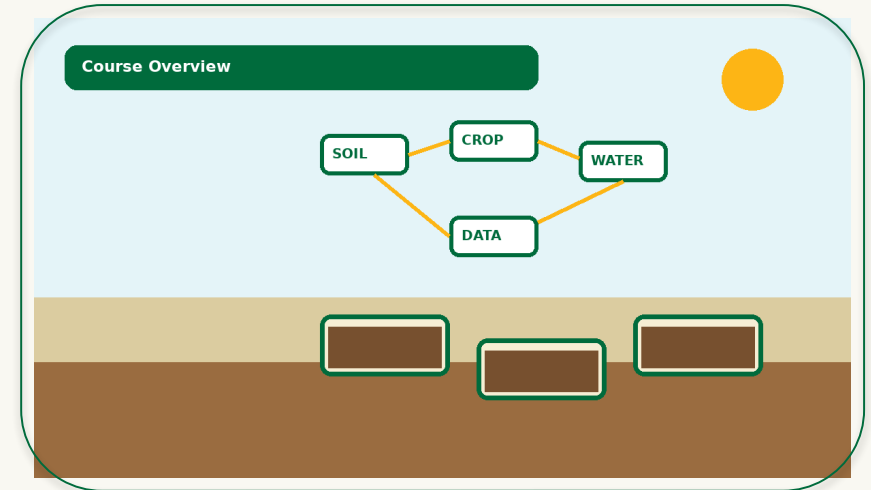
- LIFEWS AgriShine™ GrowMeal™ JSS 1-3 Teacher Training
- Audience: JSS 1-3 teachers, agricultural-science educators and garden facilitators
- Format: Self-paced online course + supervised practical/laboratory orientation
- Recommended duration: 3.5 hours online + 2 hours practical
- JSS learning progression: formulate -> test -> measure -> analyze -> manage



Course Overview

Learning outcomes, materials and teacher readiness

- Learning outcomes: explain the garden/laboratory system; teach all ten technical topics accurately; guide students in measurement, comparison and data collection; connect agricultural science with mathematics and systems reasoning; supervise practical work safely; maintain credible records; interpret evidence without overstating conclusions.
- Core materials: teacher guide, system/plot map, approved tools and irrigation equipment, labels, clipboards, rulers/tapes, graduated containers, meters/sensors where relevant, data sheets, clean sample/harvest containers, PPE/handwashing supplies and first-aid access.
- Completion expectation: complete all modules, demonstrate one safe practical session, and submit a lesson, measurement protocol, experiment or systems investigation appropriate to the course.



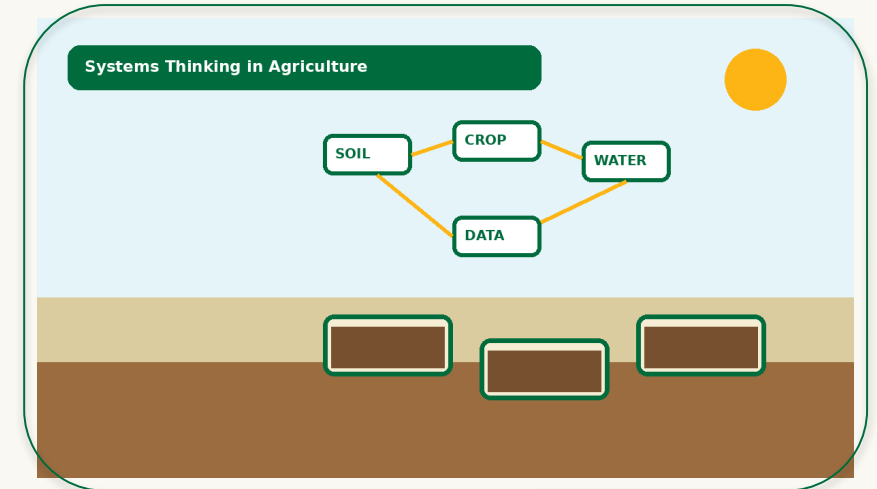
TEACHER PRACTICE

- Training sequence: QUESTION -> PLAN -> IMPLEMENT -> MEASURE -> ANALYZE -> DECIDE.

Systems Thinking in Agriculture

Teacher knowledge, technical practice and student supervision

- Define a system as interacting components organized within a boundary for a purpose.
- Identify crop, soil, water, nutrients, energy, infrastructure, people and information as possible agricultural system components.
- Focus on relationships and feedbacks rather than studying each component in isolation.
- Use causal or flow diagrams appropriate to JSS level.
- Recognize that changing one component can create effects elsewhere.



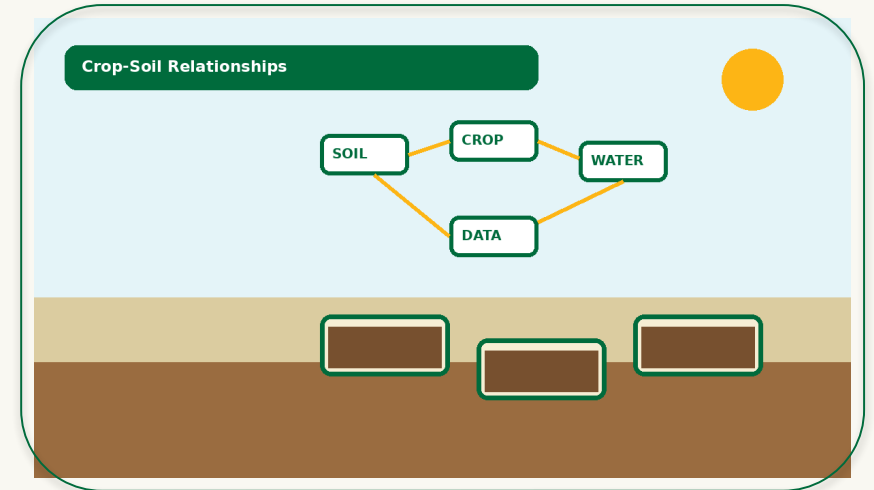
TEACHER PRACTICE

- Map the main components and connections in the school garden.

Crop-Soil Relationships

Teacher knowledge, technical practice and student supervision

- Trace how soil supports roots, stores water, supplies nutrients and influences crop conditions.
- Show how crop roots and residues can also affect soil structure and organic inputs.
- Use measurements such as moisture, infiltration or crop growth to examine relationships.
- Avoid treating soil as a passive container.
- Connect soil management decisions to crop response.



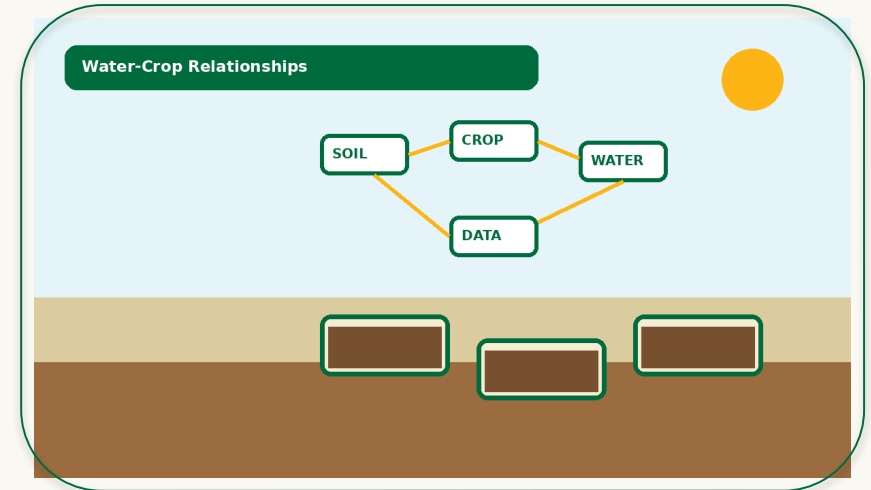
TEACHER PRACTICE

- Prepare a crop-soil interaction diagram with two measurable indicators.

Water-Crop Relationships

Teacher knowledge, technical practice and student supervision

- Connect water availability, root-zone moisture, crop stage and growth.
- Identify pathways from water source to storage, irrigation system, soil/root zone and crop.
- Recognize losses through leakage, runoff, drainage and evaporation.
- Use water-use and crop records to evaluate management.
- Discuss both water shortage and excess.



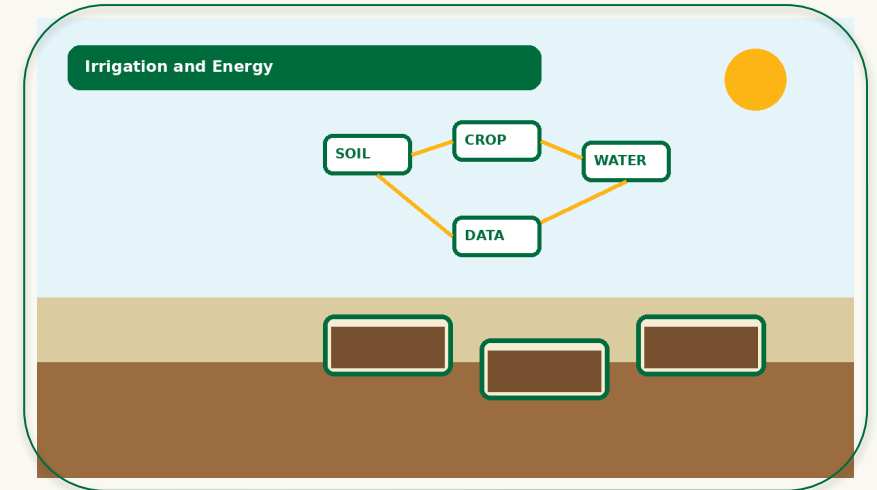
TEACHER PRACTICE

- Trace and quantify one water pathway in the garden.

Irrigation and Energy

Teacher knowledge, technical practice and student supervision

- Explain that moving or pressurizing irrigation water can require energy when pumps are used, while gravity systems may reduce pumping requirements.
- Identify pump, power source, storage and irrigation components where present.
- Relate irrigation duration and pump power conceptually to energy use.
- Keep students away from electrical panels, mains wiring and pump servicing.
- Compare management choices such as pumping to storage versus direct irrigation where the system permits.



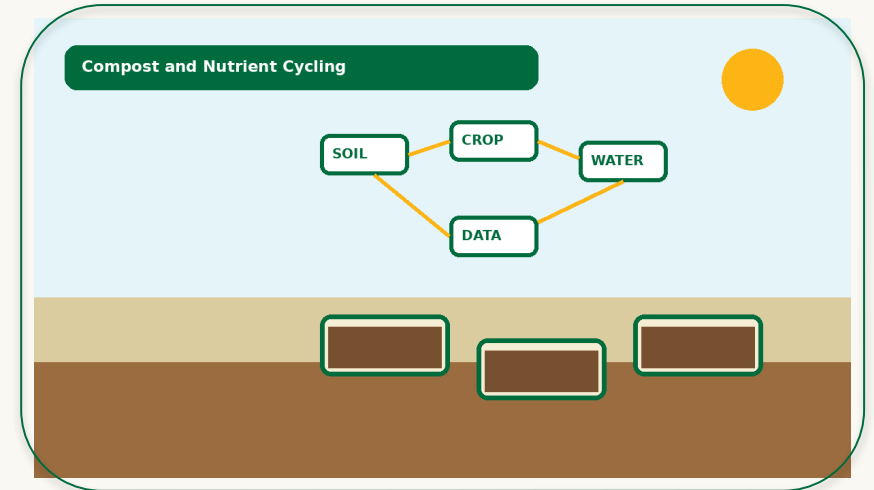
TEACHER PRACTICE

- Draw the irrigation-energy pathway and identify decision points.

Compost and Nutrient Cycling

Teacher knowledge, technical practice and student supervision

- Trace suitable organic residues from crop production to composting and mature compost application.
- Explain nutrient cycling as movement and transformation of nutrients through system components.
- Recognize that harvested produce exports nutrients from the garden system.
- Keep unsafe wastes and fresh manure outside pupil compost handling.
- Use records to estimate where organic materials enter and leave.



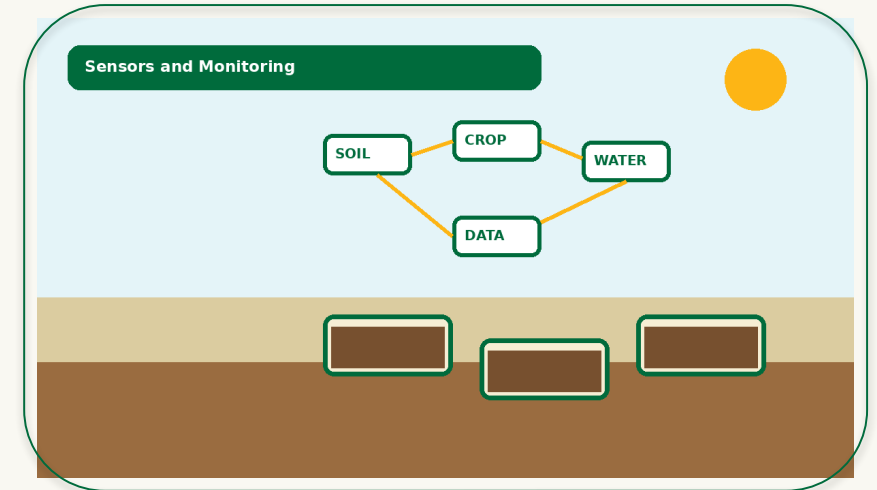
TEACHER PRACTICE

- Create a crop-residue-compost-soil-crop cycle diagram.

Sensors and Monitoring

Teacher knowledge, technical practice and student supervision

- Place sensors within the system map according to the variable they observe.
- Distinguish measurement, information and management action.
- Combine sensor data with manual observations and records.
- Identify what remains unmeasured and how that limits conclusions.
- Use monitoring to detect change, not to automate decisions without rules and supervision.



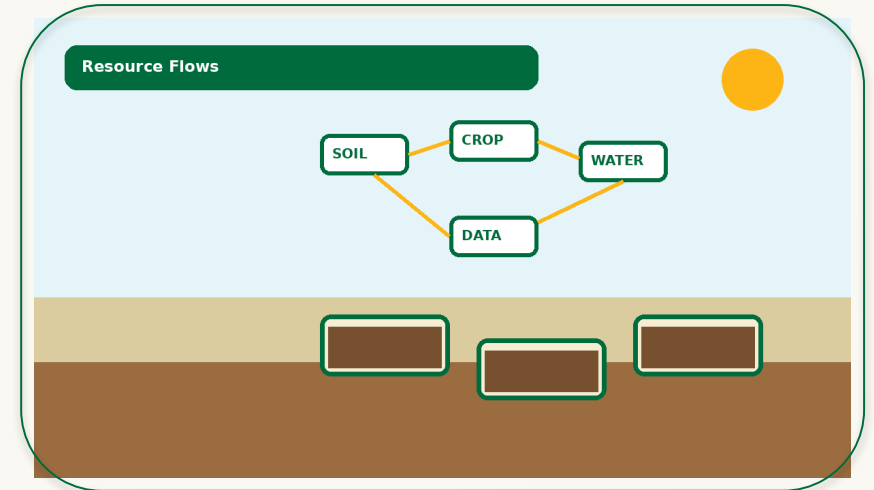
TEACHER PRACTICE

- Add monitoring points to the garden system map.

Resource Flows

Teacher knowledge, technical practice and student supervision

- Map flows of water, nutrients, crops/biomass, energy and information.
- Use arrows with clear direction and labels.
- Identify storage points such as tanks, soil moisture, compost bays or records.
- Distinguish physical flows from information flows.
- Look for losses, bottlenecks and reuse opportunities.



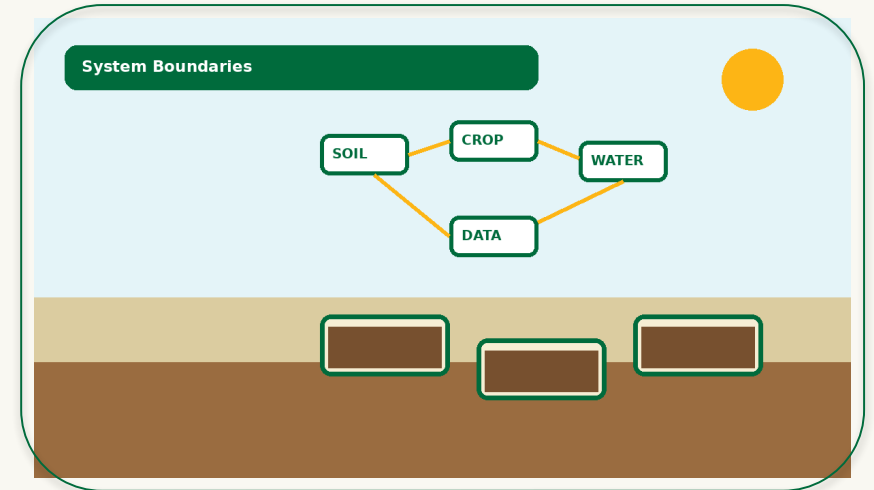
TEACHER PRACTICE

- Prepare a multi-resource flow map.

System Boundaries

Teacher knowledge, technical practice and student supervision

- Define what is inside the system being studied and what is outside but interacts with it.
- Choose boundaries based on the investigation question.
- Examples may include one bed, the entire school garden or garden plus water-storage system.
- Recognize that conclusions can change when boundaries change.
- State the boundary explicitly in student projects.



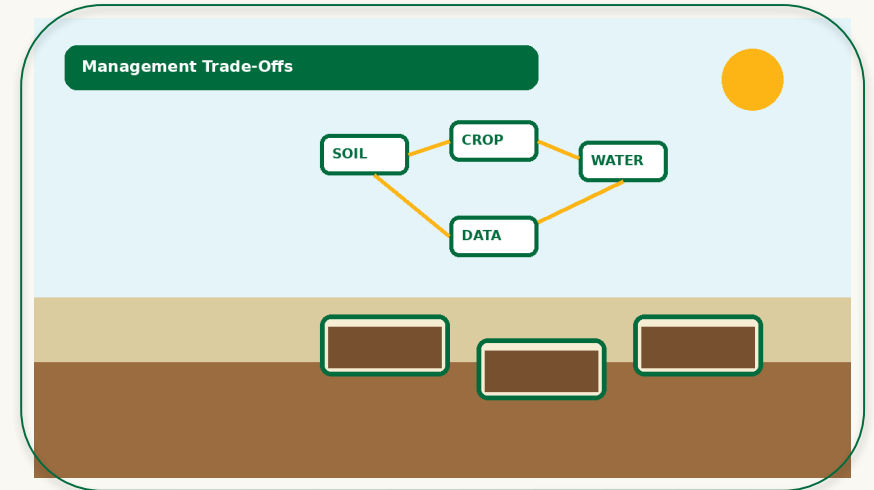
TEACHER PRACTICE

- Compare two possible boundaries for the same garden problem.

Management Trade-Offs

Teacher knowledge, technical practice and student supervision

- Define trade-offs as situations where improving one objective may reduce another or require additional resources.
- Examples include maximizing planted area versus maintaining access, or reducing irrigation versus protecting crop yield.
- Use evidence and multiple performance indicators.
- Avoid assuming one decision is optimal for every objective.
- Include safety, labor and maintenance in management evaluation.



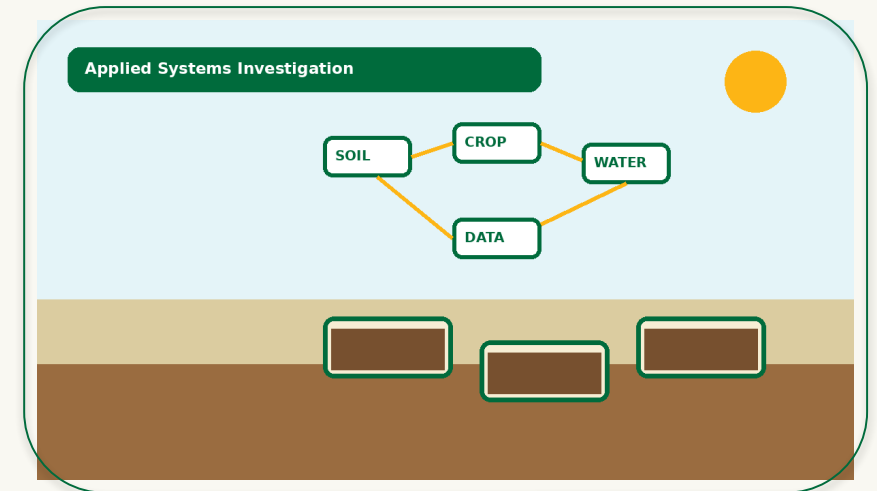
TEACHER PRACTICE

- Evaluate one garden decision using at least three criteria.

Applied Systems Investigation

Teacher knowledge, technical practice and student supervision

- Begin with a system-level question involving at least two connected subsystems.
- Map components, boundary, resource/information flows and measurements.
- Collect data using a defined protocol.
- Analyze interactions and identify a practical management option.
- State uncertainty, limitations and what should be measured next.



TEACHER PRACTICE

- Completion task: conduct a JSS applied-systems investigation from map to recommendation.